

JAMES ERNEST TARIMAN WOLFE

Full-Stack Developer / Data Engineering / Web Scraping

City Homes, Basak, Lapu-Lapu City, Cebu

Email: jameswolfe059@gmail.com | **LinkedIn:** James Wolfe | **GitHub:** james-wolfe-04

SUMMARY

Full-stack developer with production experience across the whole data stack: Python scraping pipelines, NestJS/TypeScript APIs, PostgreSQL schema design, and Next.js analytics dashboards. Ships features end to end — database migration through live UI — in a multi-repository production codebase, in a code-reviewed, CI-gated team workflow. AWS Academy certified in Cloud Architecting.

EXPERIENCE

Metawatt LLC | *Full-Stack Developer* | May 2026 – Present

- **Full-Stack Feature Ownership:** Delivered the scraper observability system end to end — the `scrape_runs` **PostgreSQL** schema on **Supabase** (row-level security, indexed read paths), the **NestJS** intake and aggregation endpoints, the **Python** reporting client, and the **Next.js** dashboard that visualizes it.
- **Backend API Development:** Built and hardened REST endpoints in **NestJS** and **TypeScript** against Supabase — bounded query parameters, TTL caching, API-key intake guards, and correct state-code mapping across all 50 states.
- **Analytics Dashboards:** Shipped the Web Scraping Trends and Breakdowns modules in **Next.js**, **React**, and **TypeScript** with **Recharts** — run volume, coverage by state, failure reasons, duplicate rates, and county run status, all wired to live production data.
- **Scraping Infrastructure:** Integrated **Crawl4AI** and containerized crawling into the Python pipeline with **Docker**, and migrated county scrapers onto the backend intake API with retry tallies and webhook delivery tracking.
- **Data Extraction:** Built and maintained county-level scrapers (Montgomery AL, Benton AR, Richland OH, Clay MO) with **Selenium**, **Playwright**, and PDF/OCR extraction (**pdfplumber**, **PyMuPDF**, **Tesseract**) inside a shared pipeline covering 25+ jurisdictions across 9 states.
- **Code Quality & Collaboration:** Worked a Git-flow pull-request process with a distributed team across four production repositories, resolving review feedback and maintaining CI lint gates (**Ruff**, **ESLint**, **Prettier**).

Metawatt LLC | *Web Scraping / Data Analytics Intern* | Jan 2026 – April 2026

- **Automated Data Pipelines:** Developed and maintained complex web scraping pipelines using **Python**, **Selenium**, **Playwright**, and **BeautifulSoup** to extract high-volume lead analytics and surplus fund data.
- **Jurisdictional Data Extraction:** Built specialized scrapers for multiple state and county-level reports, ensuring high data accuracy for lead volume identification.
- **Serialization & Optimization:** Implemented custom serialization logic to organize extracted data into structured local directories, improving data retrieval efficiency for the analytics team.
- **Agile Collaboration:** Worked closely under a Team Leader to deliver technical assignments, manage GitHub repositories, and present progress on the "Excess Elite Lead Analytics" project.

PROJECTS

Surplus Funds Data Platform | *Full-Stack Developer (Metawatt)* | 2026

- End-to-end lead pipeline: **Python** scrapers extract county tax-sale surplus records (parcel IDs, ownership details, claim deadlines), report run telemetry to a **NestJS** API, and land in **Supabase PostgreSQL** for the analytics dashboard.
- Designed the `scrape_runs` table and its indexed read paths, then built the intake endpoints and the **Next.js** / **Recharts** dashboard modules that surface scraper health, coverage, and failure reasons.
- Handled dynamic pages, PDF-only sources, and bot protection using **Playwright**, **Crawl4AI**, OCR, and stealth drivers to keep data flowing across inconsistent regional site structures.

ETEEAPP | *Project Manager* | 2025

- **Live:** eteeapp.netlify.app
- Led a development team at Cebu Institute of Technology – University to build a functional web application.
- Contributed heavily to backend business logic and frontend architectural structure.

SleepSync - Wellness App | *Mobile Developer* | 2024

- **Live:** sleepsyncapp.netlify.app
- Developed a wellness platform focused on sleep quality optimization.
- Engineered the mobile version using **Android Studio** to ensure a seamless user experience.

SKILLS

- **Languages:** Python, TypeScript, JavaScript, Java, Kotlin, SQL (PostgreSQL, MySQL), HTML, CSS
- **Backend:** NestJS, Node.js, Spring Boot, RESTful API design, Supabase
- **Frontend:** Next.js, React, TypeScript, Tailwind CSS, Recharts, TanStack Table
- **Data Engineering:** Web Scraping (Selenium, Playwright, BeautifulSoup, Crawl4AI), PDF/OCR Extraction (pdfplumber, PyMuPDF, Tesseract), Automation Pipelines, Data Serialization
- **Databases:** PostgreSQL / Supabase (schema design, migrations, RPCs, row-level security), MySQL
- **Tools & Platforms:** Git, GitHub (Git-flow, PR review), Docker, AWS Cloud, CI/CD (GitHub Actions), Ruff / ESLint / Prettier, Linux/Terminal environments
- **Mobile Development:** Android Studio (Java/Kotlin)

EDUCATION

Cebu Institute of Technology – University | 2021 – 2026

- Bachelor of Science in Information Technology (Graduated 2026)

Indiana Aerospace University | 2018 – 2020

- *Senior High School: STEM (Science, Technology, Engineering, Mathematics)*

CERTIFICATIONS

- **AWS Academy Graduate:** Cloud Architecting
- **AWS Academy Graduate:** Cloud Foundations
- **Kaggle:** Data Visualization